MURMURATIONS OF CM ELLIPTIC CURVE FAMILIES: AN EXPERIMENTAL STUDY WITH CONTROLS

ABSTRACT. We study the murmuration phenomenon—the anti-phase correlation between Frobenius traces of rank-0 and rank-1 elliptic curves—restricted to families with complex multiplication by a single imaginary quadratic field. Using exact ranks and fixed conductor windows, with shuffle controls that isolate rank from conductor, we establish that single-field CM families exhibit genuine murmurations ($z = -6.8$ for $\mathbb{Q}(\sqrt{-3})$, $z = -5.5$ for $\mathbb{Q}(i)$). The CM murmuration density is supported *exactly* on the primes that split in the CM field and is identically zero on inert primes, a structural feature with no generic analogue. A matched-sample comparison finds suggestive but not statistically significant evidence ($p \approx 0.13$) that CM families murmur with larger amplitude than generic families at equal sample size. We document a sequence of controls that refuted several natural but incorrect hypotheses along the way, and we pose the derivation of a closed-form CM murmuration density—via the CM specialization of the Eichler–Selberg trace formula—as the central open problem, for which we prove the support structure and sketch the route.

1. BACKGROUND

For an elliptic curve $E/\mathbb{Q}$ of conductor N and a prime $p \nmid N$, let $a_p(E) = p + 1 - \#E(\mathbb{F}_p)$. The *murmuration* phenomenon, discovered by He–Lee–Oliver–Pozdnyakov, is an oscillating correlation between a_p and the rank (equivalently the sign of the functional equation) when curves are averaged over a conductor window and separated by rank. Zubrilina gave a closed-form density for the analogous phenomenon for holomorphic modular forms via the Eichler–Selberg trace formula; a closed form for the elliptic-curve case remains open. We ask what the CM case looks like, where a_p is rigidly governed by a Hecke character.

2. METHOD AND CONTROLS

We draw curves from the Cremona database with exact analytic rank ($r \in \{0, 1\}$), tagged by whether they have CM and, if so, by the CM discriminant. For a single CM field with discriminant D_K, a prime p of good reduction is *inert* iff $\left(\frac{D_K}{p}\right) = -1$, and then $a_p = 0$; otherwise p splits and $a_p = \pi + \bar{\pi}$ for a generator π of the prime above p. The murmuration statistic is the correlation between the family-averaged profiles $\langle a_p/\sqrt{p} \rangle$ of the two rank classes.

Remark 1 (Controls that mattered)**.** Three design errors were caught and corrected during this study, each by a control; we record them because the corrected conclusions depend on them. *(i)* A mixed-CM-field family conflates the distinct splitting laws of the different CM fields; the split/inert decomposition is only meaningful for a single field. *(ii)* Selecting the first N curves of each rank separately gives *different conductor windows* per rank, contaminating the murmuration with a conductor-range effect; we fix a common window and shuffle within it, so the null model isolates rank. *(iii)* Comparing a CM family of ~ 100 curves to a generic family of ~ 30000 conflates amplitude with measurement precision ($\sigma_{\text{mean}} \propto N^{-1/2}$); the honest comparison equalizes sample size. Each correction reversed a tempting conclusion.

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3. Results

Proposition 2 (Single-field CM murmurations)**.** *In a fixed conductor window with rank isolated by within-window shuffling, families with CM by* $\mathbb{Q}(\sqrt{-3})$ *and by* $\mathbb{Q}(i)$ *exhibit genuine murmurations, with anti-phase correlations* -0.28 *(*$z = -6.8$*) and* -0.38 *(*$z = -5.5$*) respectively on split primes, against shuffle baselines near zero.*

Theorem 3 (Support on split primes)**.** *For a family with CM by an imaginary quadratic field* K*, the murmuration density* $D_{\mathrm{CM}}(p)$ *is supported on the primes that split in* K *and is identically zero on inert primes:*

$$p \text{ inert in } K \implies a_p = 0 \text{ for every curve} \implies D_{\mathrm{CM}}(p) = 0.$$

Verified to machine precision (rank-0 *mean on inert primes* $= 0.00000$ *for both* $\mathbb{Q}(\sqrt{-3})$ *and* $\mathbb{Q}(i)$*). This has no generic analogue: generic families are supersingular at a density-zero set of primes.*

Observation 4 (Amplitude, suggestive but not significant)**.** At matched sample size $N = 100$ per rank in a common conductor window, the generic murmuration amplitude is -0.144 ± 0.060 (over 100 subsamples), while the CM value is -0.276. Treated as a comparison of two distributions of comparable sampling error, the difference is $z \approx 1.6$ ($p \approx 0.13$): suggestive that CM families murmur with larger amplitude at fixed N, but *not* statistically significant. The much larger full-generic amplitude (-0.90) reflects sample size, not intrinsic amplitude.

Remark 5. Observation 4 is the one result pointing toward a CM-specific effect that survived every control. We flag that our first significance estimate ($z = -2.2$) was inflated by comparing the CM value as a fixed point against the generic distribution; the fair distribution-vs-distribution estimate is $z \approx 1.6$. Establishing the effect requires bootstrapping the CM family, larger N (beyond the Cremona range, via LMFDB), and replication across CM fields.

4. The closed-form question

The central open problem is whether the rigidity of CM Frobenius traces yields a closed-form murmuration density, where the generic case is unsolved.

Conjecture 6 (Closed-form CM density)**.** *For a family with CM by* K*, attached to Hecke Grössencharacters* ψ*, the murmuration density admits a closed form, supported on split primes, expressible through Hurwitz–Kronecker class-number sums* $H(4p - t^2)$ *subject to the CM condition, obtained by specializing the Eichler–Selberg trace formula.*

Route (sketch, not a proof). For CM forms $a_p = \psi(p) + \overline{\psi(p)}$, so the family sum $\sum_E a_p(E)\, w(E)$ defining the murmuration becomes a sum over Hecke characters, a theta/Eisenstein-type sum. The Eichler–Selberg trace formula, applied to the CM family, reduces the trace of Hecke operators to class-number sums; the CM condition restricts the t-sum to those t with $t^2 - 4p$ in the correct square class. The support on split primes (Theorem 3) and the CM Sato–Tate law (the angle θ_p, $a_p = 2\sqrt{p}\cos\theta_p$, equidistributes uniformly on split primes) are the two structural inputs. We emphasize that we have *not* carried out this derivation; we establish its support structure and observe numerically that the split-prime density is oscillatory rather than a monotone decay, consistent with a nontrivial closed form.

Remark 7 (Scope)**.** We claim: single-field CM murmurations exist (Prop. 2); the density is supported on split primes (Thm. 3, proven); the amplitude may exceed generic at fixed N (Obs. 4, $p \approx 0.13$, not significant); and a closed form is conjectured with a concrete route (Conj. 6, not proven). We explicitly do not claim to have derived the closed form, nor that

CM is “special” beyond the proven support structure. The murmuration phenomenon itself is universal; CM families participate in it with the additional rigidity that their density vanishes on inert primes.

References

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